Water filters: HMTc Standard

Karen Pendergrass · · Heavy Metal Tested & Certified, Institute of Contaminant Standards

CITE AS

Pendergrass K. "Water filters: HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.com/standards/water-filters. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for water filters in Food-Contact Consumer Goods and Kitchenware, evaluated on nET ELEMENT RELEASE FROM THE TREATMENT DEVICE INTO THE TREATED WATER, µg/L reported as µg/kg, determined as device effluent minus a matched influent blank so that any contribution BY the device is resolved separately from what the device removes. This is a device-release basis on the water leaving the device; the sovereign drinking-water parametric values are a WATER basis on the supply as delivered and are disclosed as companions, never republished as the device's release ceiling.. No government sets a numeric heavy-metal limit for this product form, so every panel analyte is held under mandatory analytical control or a substantiated not-material determination rather than an invented ceiling.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum

Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	control	measured every lot; no human-food maximum
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION

Lead (Pb)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Lead is the element a filter is most often bought to remove, and it is simultaneously a possible device contribution from leaded brass fittings, bronze bodies and soldered joints in the wetted path. Both directions are tested: net contribution against a matched influent blank, and removal performance against the challenge influent. The sovereign drinking-water parametric value disclosed as this cell's companion is a limit on the WATER as delivered; it is not the device's release ceiling and is not published as one. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Cadmium (Cd)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Cadmium can be contributed by cadmium-plated or cadmium-brazed wetted components and is a target of reduction claims. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Total arsenic (tAs)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Arsenic is a principal reduction target for point-of-use devices and an impurity route in copper-alloy wetted parts. Determined as the total element. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Inorganic arsenic (iAs)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Inorganic arsenic is the species that matters in water and is essentially the whole of the arsenic present, so it is determined in its own right rather than bounded: this row's total-arsenic cell is itself a control with no sovereign device ceiling, so no published parent could bound the species. The US EPA arsenic maximum contaminant level disclosed as this cell's companion is a WATER-basis limit on the treated supply. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Mercury, total (tHg)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Mercury is a reduction target for some media and has no functional role in the wetted path; determined as the total element. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Methylmercury (MeHg)	control	Methylmercury on this row FAILS SAFE TO A CONTROL rather than to a not-material determination, and the reason is specific to this material family. Total mercury on this row is itself a control with no published number, so no published parent bounds the species; a not-material determination here would rest entirely on the pathway argument, with no numeric backstop. That argument is weaker on this row than anywhere else in this category. A granular or block activated-carbon bed, an ion-exchange resin bed and a wetted mineral or ceramic medium are permanently wetted, carbon-rich and colonisable, and this row's own conditioning, rated-capacity and change-out requirements concede that the media change biologically and temporally in service. A biofilm-colonised bed operating at or beyond rated capacity, holding the inorganic mercury it has removed, is the one configuration in Category 23 where an anaerobic microbial methylation niche is physically available. HMTc therefore does not assert that methylmercury is immaterial here; it mandates the determination instead, with no number invented and no sovereign parametric value republished as a device release ceiling. Directive (EU) 2020/2184 Annex I and the US EPA National Primary Drinking Water Regulations both state mercury as the TOTAL element, so no sovereign methylmercury value exists to anchor even if one were wanted, and methylmercury never substitutes for the total-mercury determination.

Nickel (Ni)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Nickel is contributed by nickel-plated and stainless wetted fittings and by some KDF-type redox alloy media, which are copper-zinc alloys whose impurity profile must be verified. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Tin (Sn)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Tin can be contributed by tin-bearing solder in the wetted path. No sovereign drinking-water parametric value for tin exists in the credible set, so this cell carries no companion at all and is a pure control.

Aluminium (Al)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Aluminium is a device-contribution risk from activated-alumina arsenic-selective media, from alum-containing media and from aluminium housings, and it is the one element a metal-removal device can readily ADD; the net-contribution test against the matched influent blank is the operative determination. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Total chromium (Cr)	control	A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling. Total chromium is retained as a screening determination for stainless and plated wetted parts; the health-relevant determination is the hexavalent-chromium cell, because chromium(III) in water is the benign species. The two mandatory determinations on this row are net device contribution against a matched influent blank and reduction performance at end of rated life; either one alone misrepresents the device.

Hexavalent chromium (Cr-VI)	control	Hexavalent chromium is the health-relevant chromium species, is a named reduction target for anion-exchange and reverse-osmosis devices, and can be contributed by chromate-passivated wetted parts. No credible sovereign sets a hexavalent-chromium release ceiling for a treatment device, and this row's total-chromium cell is itself a control, so no published parent bounds the species. The fail-safe applies, and reduction performance is verified separately from net contribution. A drinking-water treatment device is an intervention whose purpose is to REMOVE metals, so the certification question is not whether the effluent meets a food limit but whether the device itself contributes any element to the water it treats. No credible sovereign sets a release ceiling for a treatment device or its media: the EU regulates materials in contact with water intended for human consumption through the Art. 11 hygiene-requirement and positive-list route of Directive (EU) 2020/2184 rather than through element release values, the US route is the SDWA lead-free content definition plus NSF/ANSI 61 conformity, and neither is a sovereign element release ceiling in this register. Regulation (EU) 10/2011 does not reach the media at all: Art. 2(2)(a) expressly excludes ion exchange resins from its scope. The cell is therefore a control with mandatory net-contribution testing against a matched influent blank, and the sovereign drinking-water parametric value is disclosed as a companion obligation in the WATER basis, never republished as a device release ceiling.

4 | Cite this standard

Pendergrass, K. "Water filters." In Heavy Metal Tested & Certified (HMTc) Food-Contact Consumer Goods and Kitchenware standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/water-filters. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/water-filters.